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Instagram Post Engagement Rate Analysis

Abstract
Content creators and businesses rely on Instagram engagement to grow their audience and reach potential customers. It is commonly debated that the time of day a post is published affects how users interact with it. Posts published when followers are more active may receive more likes, comments, shares, and saves. This analysis investigates whether posting time, specifically morning (6-10 AM) versus evening (6-10 PM), has a statistically significant effect on Instagram post engagement rate.

Using a dataset containing Instagram post analytics, a two-sample t-test was conducted to compare mean engagement rates between the two time groups. The null hypothesis states that posting time does not affect engagement rate (H_0: $\mu_{morning} = \mu_{evening}$), while the alternative hypothesis states that posting time affects engagement rate (H_1: $\mu_{morning} \neq \mu_{evening}$). Results indicate no statistically significant difference between groups, leading to a failure to reject the null hypothesis. This finding held across stratified analyses by content category and follower count.

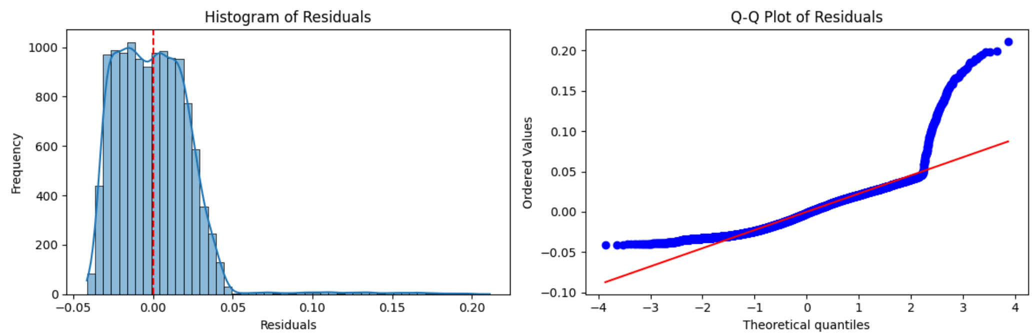
Data Facts
The dataset used in this analysis is the Instagram Analytics Dataset, publicly available on Kaggle (https://www.kaggle.com/datasets/kundanbedmutha/instagram-analytics-dataset). It contains approximately 30,000 Instagram posts.

No missing values were present in this dataset. The dataset was filtered to extract two subgroups: morning posts (post_hour ≥ 6 and ≤ 10) and evening posts (post_hour ≥ 18 and ≤ 22), which resulted in 12,500 rows of data. Accounts were further binned into four follower tiers (≤ 5,000; 5,000-10,000; 10,001-20,000; and > 20,000 followers) for stratified analysis.

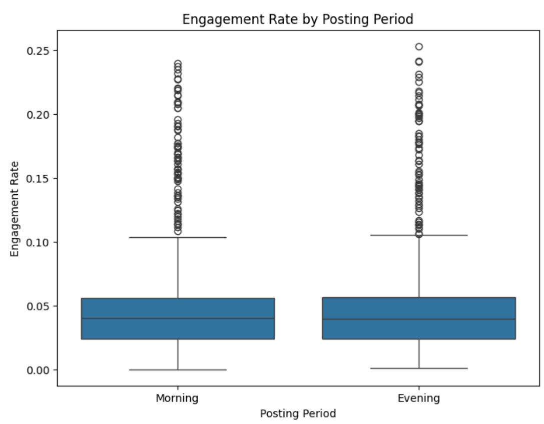
A log transformation of engagement rates was considered to address right skewness. However, visual inspection of Q-Q plots before and after transformation revealed no meaningful improvement in normality, therefore the transformation was not applied in the final analysis.

Experiment Design
The dependent variable is engagement rate, which is defined as total interactions (likes, comments, shares, and saves) divided by impressions.

A two-sample independent t-test was used to compare mean engagement rates between the morning and evening groups. Prior to testing, the normality, equal variance, and independence assumptions were assessed. After looking at the histogram of residuals and Q-Q plot, it is evident that the data is highly right-skewed. However, with approximately 6,200 posts per group, the Central Limit Theorem applies, which means the distribution is approximately normal.



Equal variance was assessed using boxplots of posting period vs. engagement rate. The boxplots of posts made in the morning and posts made in the evening were nearly identical in size and whisker lengths, so there was evidence of equal variance. To substantiate this assumption further, Levene's test was used and confirmed equal variance (p = 0.70). Therefore, Student's t-test was used in the primary analysis, where equal_var=True. For subgroup analyses, Welch's t-test was used (equal_var=False) as a default, since equal variance cannot be assumed within each subgroup without additional testing.



The independence assumption holds up because each observation in the data represents a unique Instagram post and the morning and evening periods do not overlap.

To account for potential confounders such as content category and follower count, the analysis was stratified along two dimensions:

- Content category: 10 categories (Technology, Fitness, Beauty, Music, Photography, Food, Lifestyle, Travel, Fashion, Comedy). Bonferroni-corrected threshold: $\alpha = 0.005$.
- Follower count: 4 tiers (< 5,000; 5,000-10,000; 10,001-20,000; > 20,000 followers). Bonferroni-corrected threshold: $\alpha = 0.0125$.
- Combined (content × follower): 40 subgroup combinations. Bonferroni-corrected threshold: $\alpha = 0.00125$.

Experiment Results

In the primary analysis, the two-sample t-test revealed no statistically significant difference in engagement rate between morning and evening posts. With $t = -0.2655$ and $p = 0.7906$ at a significance level of 0.05, the decision is to fail to reject the null hypothesis. Therefore, posting time has no effect on engagement rate.

T-tests were run for each of the 10 content categories with a Bonferroni-corrected threshold of $\alpha = 0.005$. One nominally significant result was observed where the content category was photography. It had a p-value of 0.049, but it did not survive Bonferroni correction. No content category showed a significant morning vs. evening effect after correction.

T-tests were run for each follower tier with a Bonferroni-corrected threshold of $\alpha = 0.0125$. No significant results were found. The lowest p-value was 0.2115 from the group with more than 20,000 followers, but it was still not statistically significant.

Across 40 subgroup combinations with a Bonferroni-corrected threshold of $\alpha = 0.00125$, two nominally significant results were observed: Technology | >20,000 followers ($p = 0.045$) and Fashion | 5,000–10,000 followers ($p = 0.046$). Neither survived Bonferroni correction. No consistent pattern emerged across the 40 combinations.

Summary and Discussion

This study found no statistically significant effect of posting in the morning vs. the evening on Instagram post engagement rate. The primary two-sample t-test and stratified analyses consistently failed to reject the null hypothesis, before Bonferroni correction, save for a few stratified analyses, and after Bonferroni correction.

The normality assumption was technically violated. Engagement rate is right-skewed with an upper tail driven by high-performing posts. However, the large sample size ensures that the Central Limit Theorem holds, making the t-test for the primary analysis valid. For smaller stratified subgroups, results should be interpreted with caution, though all nominally significant findings failed Bonferroni correction regardless. A log transformation was tested but did not reduce skewness. Independence is apparent because each observation is a unique post and the defined morning and evening periods do not overlap. Equal variance was confirmed using both a boxplot and Levene's test.

The stratified analyses address two key confounders identified: content category and account size. Neither produced significant results after correction, strengthening the conclusion that posting time is not a meaningful driver of engagement rate in this dataset.

The research goal was met. This analysis shows evidence that morning vs. evening posting time does not significantly affect Instagram engagement rate. This finding has practical implications for content creators. Optimizing what time of day to post may be less impactful than other factors such as content quality, audience targeting, or content category, which needs further investigation.

An important limitation is that this dataset may not represent all Instagram account types equally, and the possible presence of botting could introduce noise that distorts true patterns. Future work could implement a focus on verified organic engagement metrics.